

ABRAR ZAHIN RAIHAN

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EDUCATION

Bangladesh University of Engineering and Technology

Bachelor of Engineering in Computer Science and Engineering

Nov 2022 – Present

Dhaka, Bangladesh

Notre Dame College

College

Jul 2019– Dec 2021

Dhaka, Bangladesh

St. Joseph Higher Secondary School

High School

Jan 2016– Feb 2019

Dhaka, Bangladesh

INTERESTS

NLP · Information Retrieval · Multimodal AI · AI Systems · Computer Vision · Machine Learning · Mechanistic Interpretability

PUBLICATIONS

■ Causal Localization of the English Pivot in LLaVA: Mechanistic VLM Analysis and Training-Free Multilingual Steering

2026

1st Workshop on Multilinguality in the Era of Large Language Models (MeLLM) @ ACL 2026

San Diego, CA

Accepted · Archival (To appear in ACL Anthology)

- Causally localized an English-pivot bottleneck in LLaVA-1.5-7B to layers 5–17 via logit-lens analysis and causal activation patching; training-free steering vectors improved Russian VQA by +6.5 pp and Portuguese by +4.0 pp on MMBB without fine-tuning.
- [🔗 Code](#)

■ Where Do Affordances Crystallize? Video vs. Image Self-Supervised Encoders for Embodied Perception

2026

7th Annual Embodied AI (EAI) Workshop @ CVPR 2026

Denver, CO

Accepted · Non-archival · Poster (To appear on workshop website)

- Probed 25 layer positions of ViT-L encoders (V-JEPA 2/2.1, DINOv2); video SSL crystallizes affordance signal 4 layers earlier than image SSL; top-3 attention heads achieve 5.3× compute reduction with ≤ 0.1 mAP loss.
- [🔗 Code](#)

■ When Does Visual Context Help? Adaptive Modality Gating for Efficient Multimodal QA

2026

Efficient Multimodal Question Answering (EMM-QA) Workshop @ ICML 2026

Seoul, South Korea

Under Review

- AdaGate: lightweight MLP gate that decides from text alone whether to invoke the vision encoder, cutting encoder calls to ~46.5% while preserving F1 on par with always-on vision.

■ SpatialCoT: Chain-of-Thought Spatial Grounding for Viewpoint-Robust 3D-LLMs

2026

2nd Workshop on 3D-LLM/VLA: Bridging Language, Vision and Action in 3D Environments @ CVPR 2026

Denver, CO

Under Review

- SpatialCoT: five-step geometry-grounded CoT (Localize, Orient, Enumerate, Compute, Answer) that makes 3D-LLMs robust to viewpoint rotation; reduces accuracy drop at 180° rotation from 77% to 41% on Real-3DQA.

EXPERIENCE

[SysModeler.ai](#)

Research Assistant (Remote)

November 2025 – February 2026

Canton, Michigan, United States

- **SysML Layout Algorithms** Designed and implemented layout algorithms to generate structured, readable, and visually consistent SysML diagrams
- **Retrieval-Augmented Image-to-SysML** Developed a **retrieval-augmented** image-to-SysML pipeline — given an input image, the system **retrieves** structurally similar exemplars as anchors to guide generation of semantically accurate SysML representations; grounding **multimodal output** in **retrieved references** to enforce structural consistency

[Syslabo Corp](#)

SWE Intern (Remote)

Jul 2024 – Sep 2024

Tokyo, Japan

- **Org Chart Automation** Automated creation and tracking of organizational hierarchy charts, replacing a manual Excel-based system with a full-stack MERN application
- **System Design** Designed scalable database schemas and RESTful APIs for efficient hierarchical data management
- **Outstanding Performance Recognition** Received best architecture design recognition among all interns; secured a full-time job offer
-  [Internship Project](#)

PROJECTS

[IEEE VIP Cup 2025](#) | *Multimodal AI · Computer Vision · AI Systems*

Jul 2025

- **Computer vision** and **multimodal AI** system — **cross-modal RGB-IR fusion** for real-time drone detection, tracking, and payload classification under real-world degradations (blur, noise, camera instability); Champions among 40+ international teams, presented at ICIP 2025, Anchorage, AK, USA
-  [Presentation](#)  [Paper](#)  [Code](#)
-  [Read more](#)

[Intent Based Multilingual Search Optimization](#) | *NLP · Information Retrieval · AI Systems*

Apr 2025

- NLP-driven **information retrieval** system — **intent decomposition** and **multilingual dense retrieval** with in-memory **LLM inference** via llama's C++ runtime and 4-bit quantization, running CPU-only on constrained hardware; Champions among 86 teams at the Poridhi.io × Brain Station 23 AI Engineering Hackathon.
-  [Presentation](#)  [Code](#)
-  [Read more](#)

[ASR for Low Resource Bengali Regional Dialect](#) | *NLP · Multimodal AI · Machine Learning*

Nov 2025

- **Multilingual NLP** and **multimodal AI** task — **low-resource ASR** bridging **acoustic and text modalities** across 20 regional Bangladeshi dialects to standard Bangla via fine-tuned Whisper ensembles and **LLM-augmented** data synthesis; 1st Runner-Up among 90+ teams at CUET ETE Televerse 1.0.
-  [Presentation](#)  [Code](#)
-  [Read more](#)

HONORS & AWARDS

-  [CUET ETE Televerse 1.0 : AIfication 2025 \(Datathon\) - First Runner-up](#) Worked on regional Bengali dialect speech to standard Bengali text conversion. Nov 2025
-  [IEEE VIP Cup 2025 - Champions](#) Developed multimodal RGB-IR fusion models for real-time drone detection, tracking, and payload classification in surveillance videos. Presented our solution in ICIP 2025. Jul 2025
-  [Poridhi.io and Brain Station 23 AI Engineering Hackathon - Champions](#) We built an intent-based product search with llama.cpp, optimizing for low-resource use and multilingual support. Apr 2025
-  [BUET CSE Fest Hackathon 2024 - 2nd Runner-up](#) Microservices and DevOps Category Oct 2024
-  [Therap JavaFest 2024 - Top 10 finalist](#) Worked on developing an AI coding tutor with gamified learning for kids, ranked top 10 among 1300+ participants Sep 2024
-  [Code Samurai Hackathon 2024 - 10th Place](#) Engineered a waste management system for the DNCC (as proposed by BJIT), solved algorithmic challenges, and built a web app and a React Native mobile app to implement the system Mar 2024